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Remote maintenance manual - internal and external users

The REX100 WAN is an industrial router suitable for worldwide remote access to IPv4-capable end devices (PLC, HMI, IP cameras, PC, etc.).

Ethernet equipment, such as PLC control systems, can be reached with the REX 100 industrial routers, irrespective of the manufacturer.
The REX 100 routers are systematically designed to operate in conjunction with myREX24 V2 VIRTUAL.
All programming and remote maintenance is carried out exclusively through the portal.
Data transmission takes place via an encrypted VPN tunnel on the basis of the secure OpenVPN protocol.

The myREX24 V2 VIRTUAL (on premise) is managed by Draexlmaier.



1. Setup shDIALUP

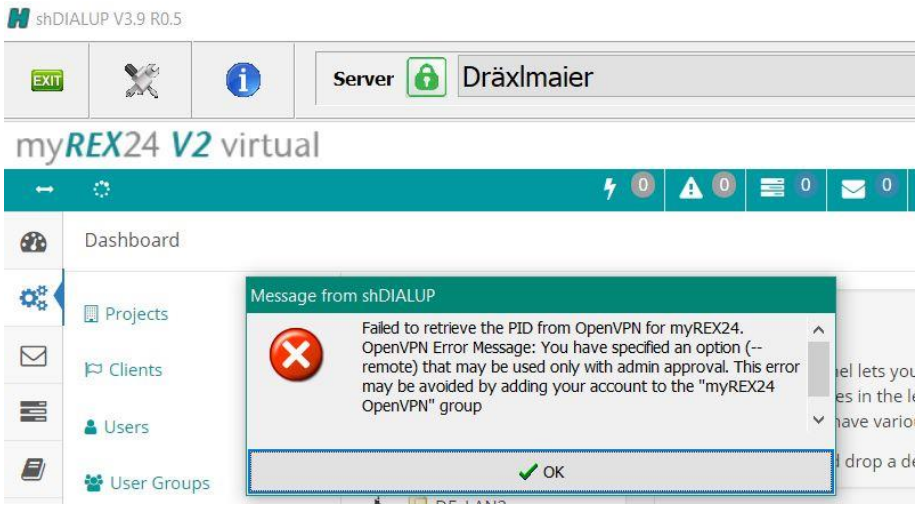
- Download the VPN client software from this link https://sales.helmholz.de/draexlmaier/
- Unzip the file
- Run the setup file as *administrator* and follow the steps.

After installing this software a new network adapter will be added to the drivers of the PC named “TAP-Windows Adapter V9”. This adapter will be used on the shDIALUP application.



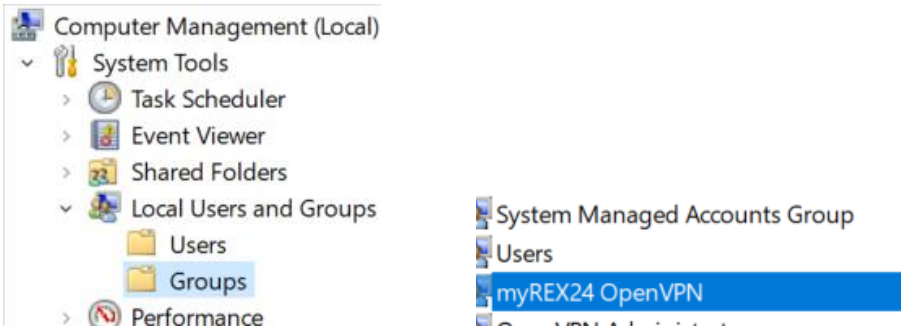
With **shDIALUP** software, an admin or user establish a connection with the server via an outbound TCP connection port 1194 (optional port 443 or 80) . The port 1194 has to be open on the local firewall and the corporate firewall.

Note: If after installation you will get the next message check if the user belong to the myREX24 OpenVPN group:



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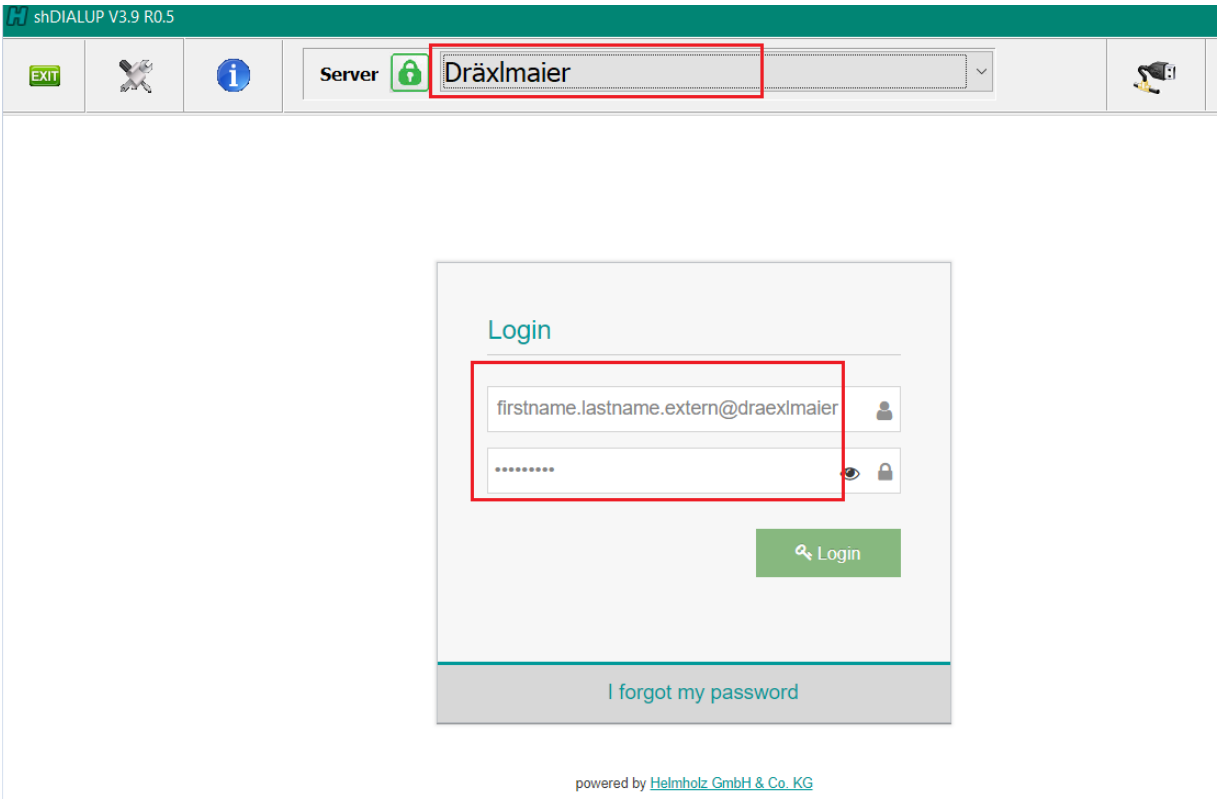
When installing the app a group myREX24 OpenVPN will be created automatically and the user should be added in this group:



1. shDIALUP login

a. The server “Draexlmaier” is selected .

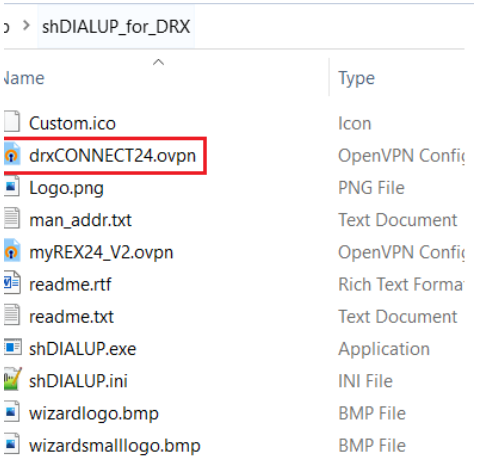
b. Enter the login credentials communicated per e-mail from “D-RemoteMaintenance@draexlmaier.com” after your account was created in portal by DRX IT (requested by the line planner)
Username should be like **firstname.lastname.extern@draexlmaier** (the user format is not applied for ANU user, this is firstname.lastname@draexlmaier same as for internal users)



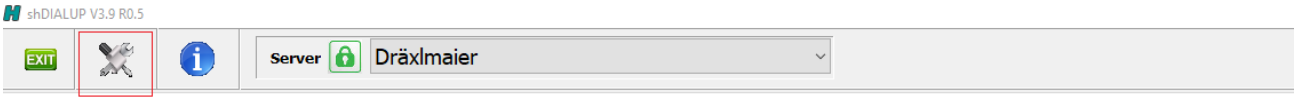
- If the app is downloaded from https://sales.helmholz.de/draexlmaier/ the DRX server settings will be automatically configured in shDIALUP.
The file drxCONNECT24.ovpn is used.

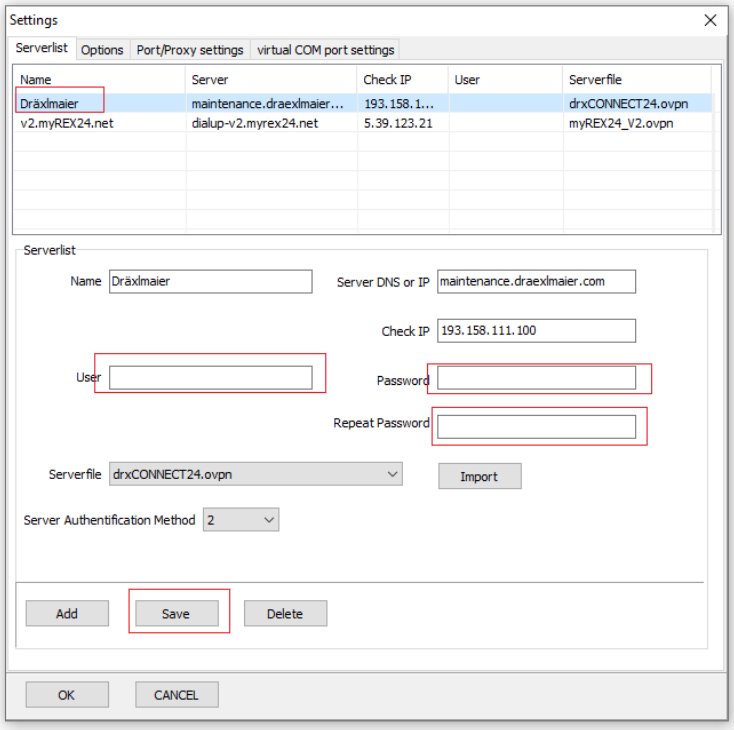
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The file is inside the app. folder:



The default server is Draexlmaier, the user and pw. should be added:





- If the external user (supplier) is already using the tool " shDIALUP " (from Helmholz) or "mbdialup" (from MB Connectline), he does not need to reinstall the tool. The Draexlmaier server can be added in his existing software.
An additional server entry for the Draexlmaier server is simply stored under the Settings-Serverlist. The user can switch between his servers via the drop-down menu.
Ovpn.file has to be downloaded and added in shDIALUP from IMPORT button. Will be automatically loaded when Drx server will be selected. The other one ovpn file used by other server will not be deleted.

All detailed information are the link: https://sales.helmholz.de/draexlmaier

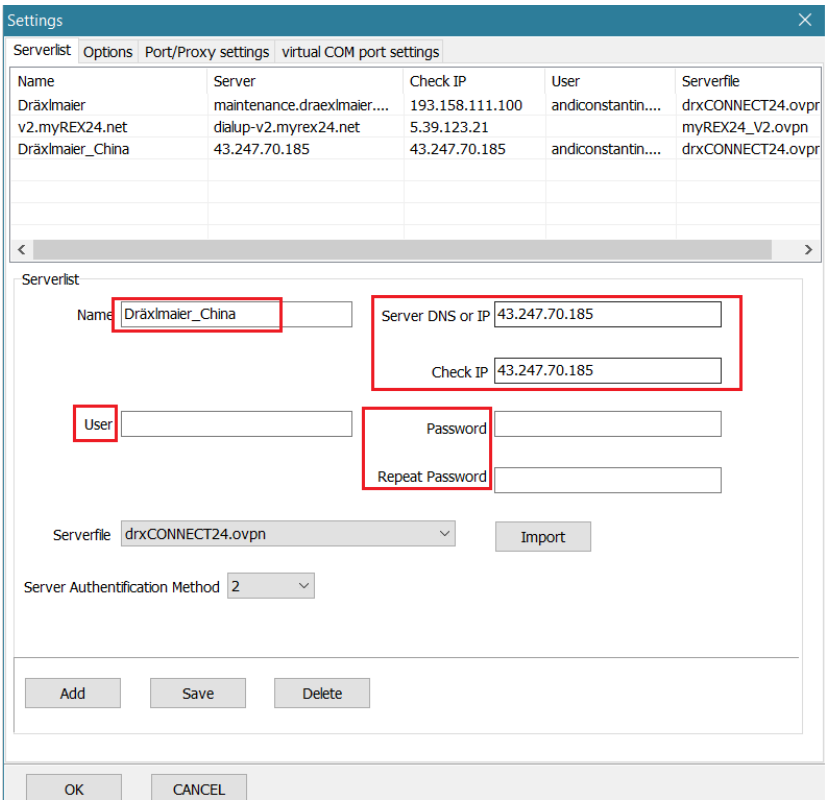
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Necessary Settings if you are using the shDialup from another location (not from China):
Name: Draexlmaier
Server DNS: maintenance.draexlmaier.com
Check IP: 193.158.111.100
Serverfile: drxCONNECT24.ovpn
Server Authentication Method: 2

shDialup used from China country

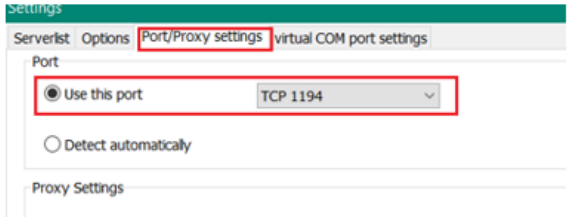
If you want to have a vpn connection from China country (inside or outside the plant) you should have the next settings in shDialup application. This is mandatory to have a stable vpn connection. shDialup app. is using a Gateway. You can modify the Draexmailer server or you can add a new one for China.

The user and pw. should be added.



Necessary Settings if you are using the shDialup from China location:
Name: Draexlmaier *or you can create a new one* ***Draexlmaier_China***
Server DNS: 43.247.70.185
Check IP: 43.247.70.185
Serverfile: drxCONNECT24.ovpn
Server Authentication Method: 2

Set the port 1194:

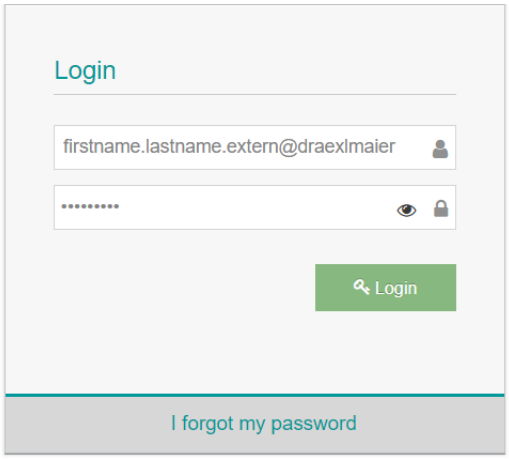


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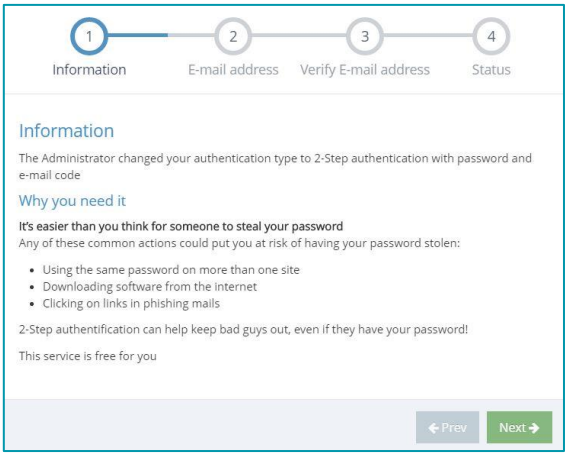
Login to the RSP (Remote Service Portal):

For the first login you will get these 4 steps:

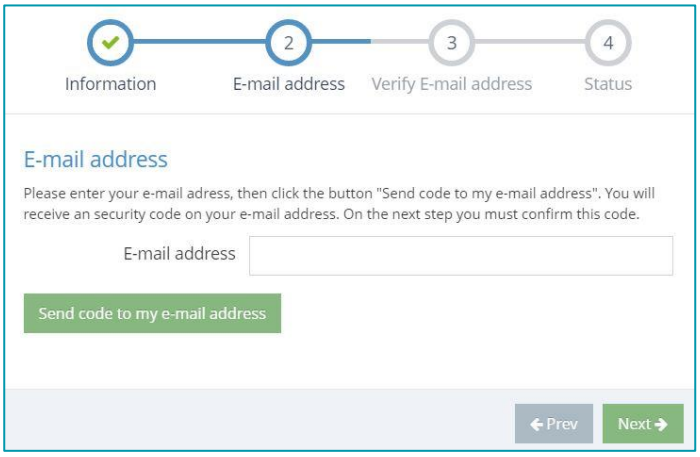
1- Enter the login credentials



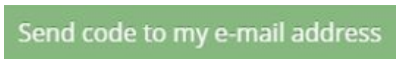
2- Next



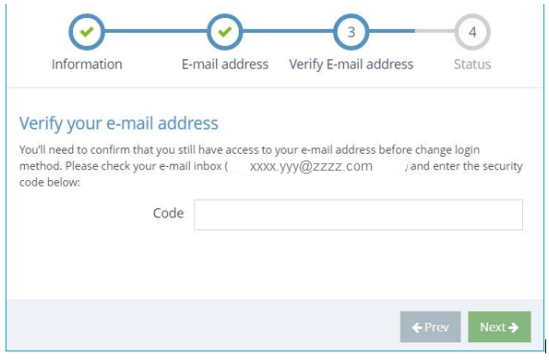
3- Enter e-mail address for which the code will be sent



and click on:



4- Enter the code already received in your e-mail



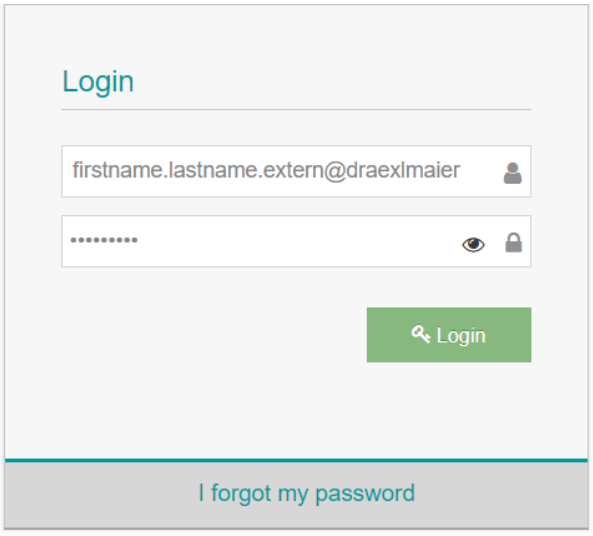
5- Confirmation



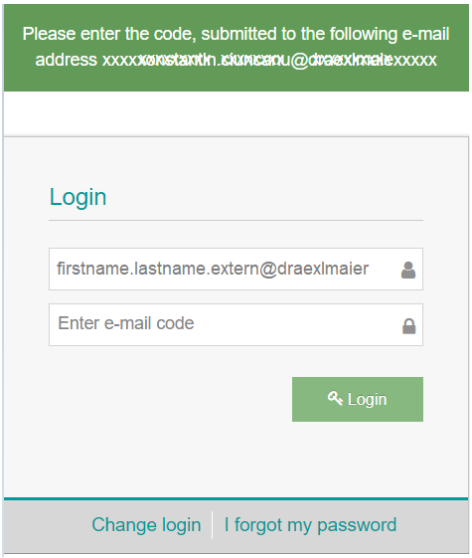
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Then for the next logins just 2 steps will be needed:

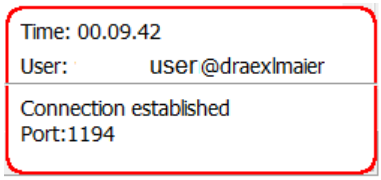
a- Password



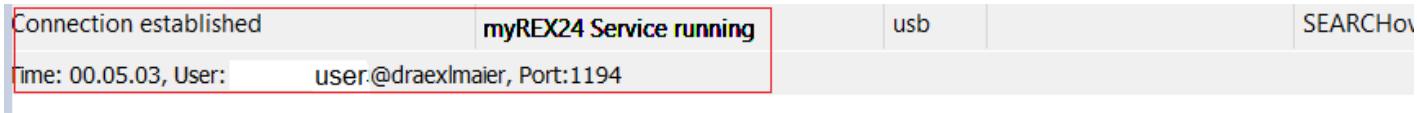
b- Code received per email



After login, in the bottom right corner will appear the information about the connection (only for shDialup downloaded from https://sales.helmholz.de/draexlmaier):



When the message "connection established" appear, the portal can be used.
Also this message can be seen on the information bar of the application:



Also the connection to the RSP (Remote Service Portal) can be done with an **HTTPS-**secured direct call from your browser.
Use the following URL: https://maintenance.draexlmaier.com

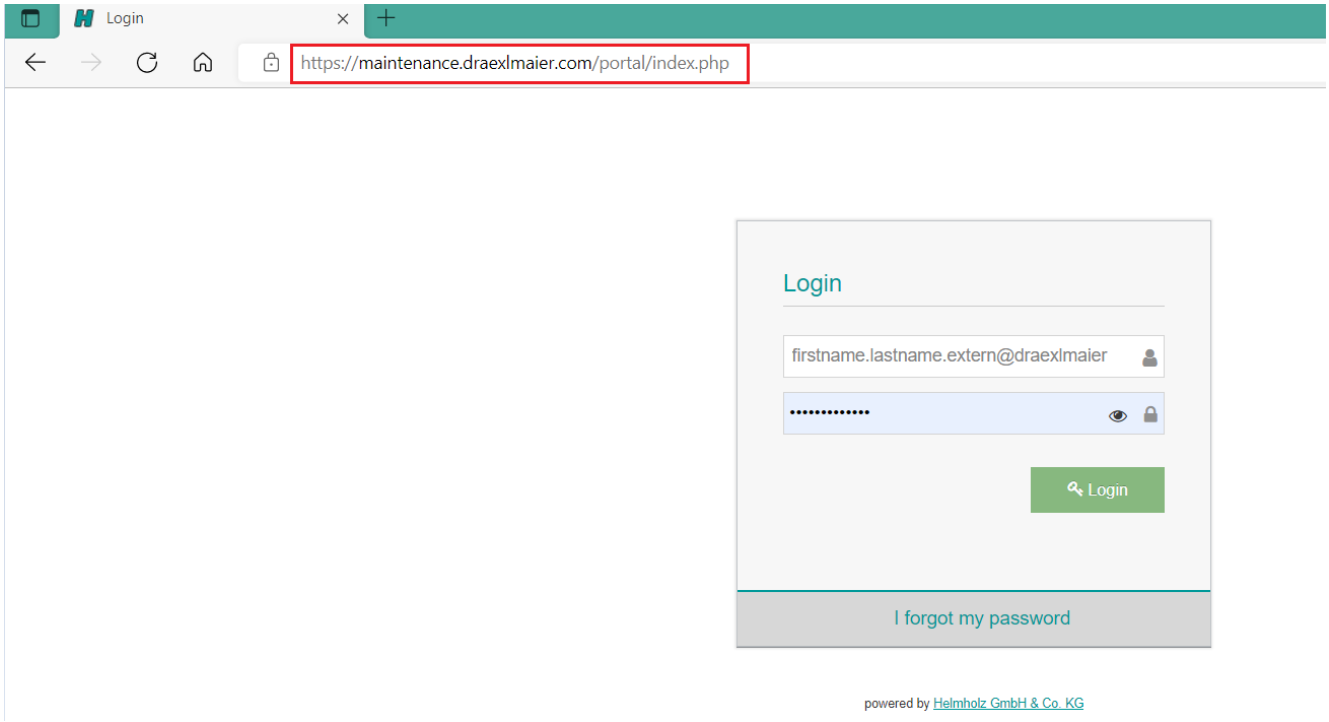
Notes:

It is not possible to start a VPN tunnel via web browser .

The feature **web2go** can be used for connection to the machine (VNC,RDP,HTTP, etc)

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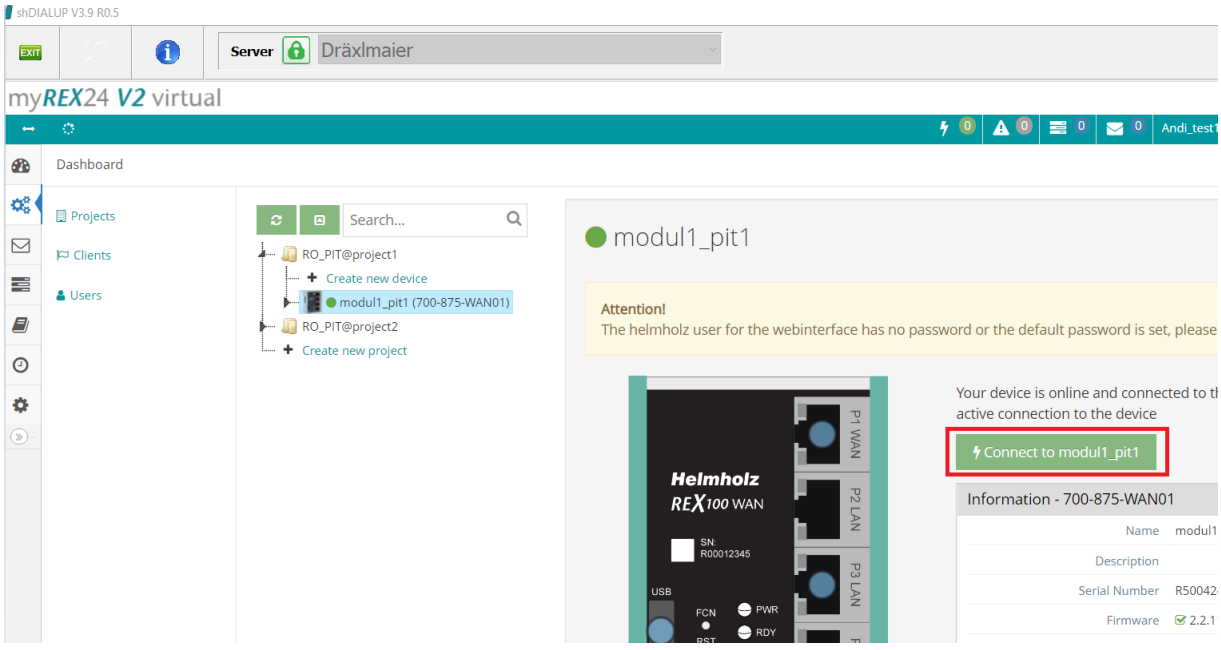
Web browser login:



2. VPN connection

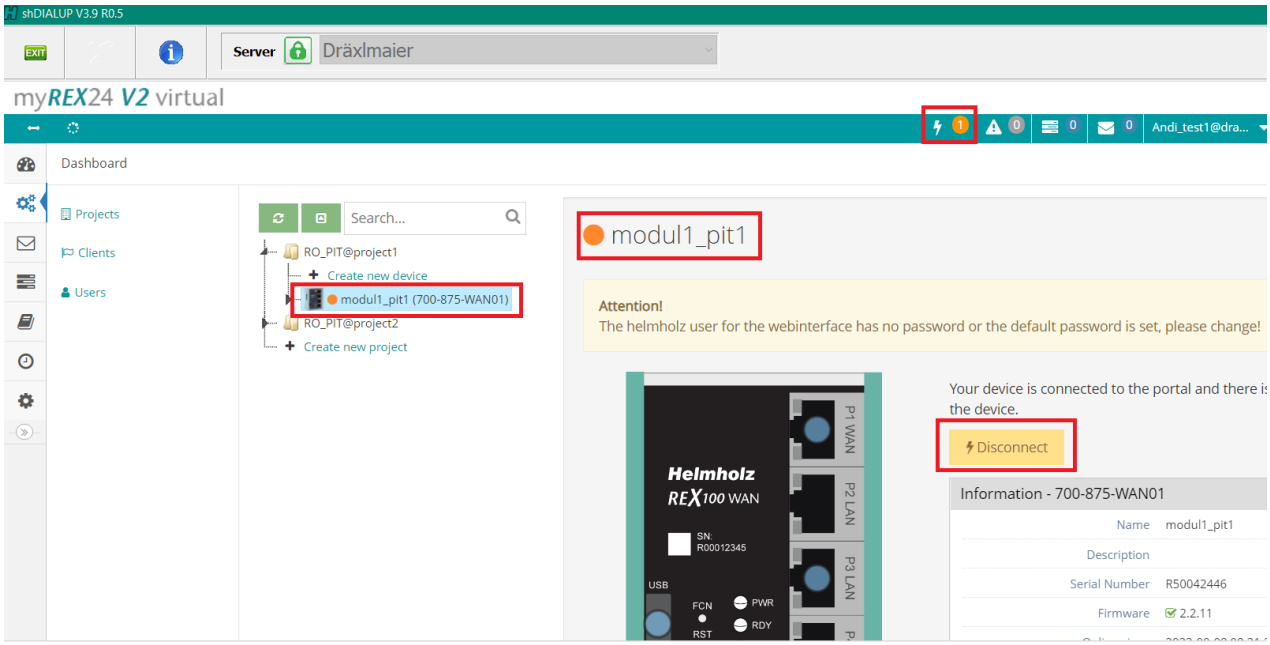
An active connection only exists when a user has established a connection to a system and activated the VPN tunnel after logging to the portal. By logging into the portal itself, there is no active connection.

In the shDIALUP app select the concerned project then the appropriate device and click "**Connect to** [machine]"



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A VPN connection is established now to the router and the green mark becomes orange:



Once the VPN connection to the REX 100 has been established, all the IPv4 packets valid for the applicable LAN network on the system side will be forwarded and you can connect on affiliated devices to the machine.

Devices as: PLC, HMI, PC, Camera, Server etc

These devices can be managed through sessions opened from your Pc with RDP, VNC, HTTPS,TIA portal etc

3.1 Multiple active connections or concurrent active connections

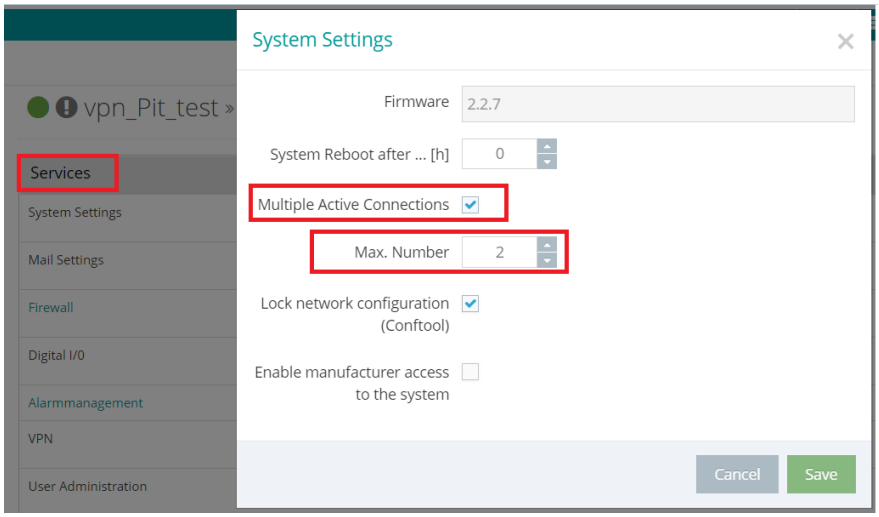
Using the remote client, multiple users can establish a VPN connection to this device simultaneously.

It means that different users can use simultaneous vpn connections to the same vpn box.

This multiple connection function is not available to users logging into the portal via a web client (https connection)

The external user have to add the information about the number of the active connections for the vpn box in the document "**Helmholz_Remote_Support_Template_DRX-accounts_and_devices**".

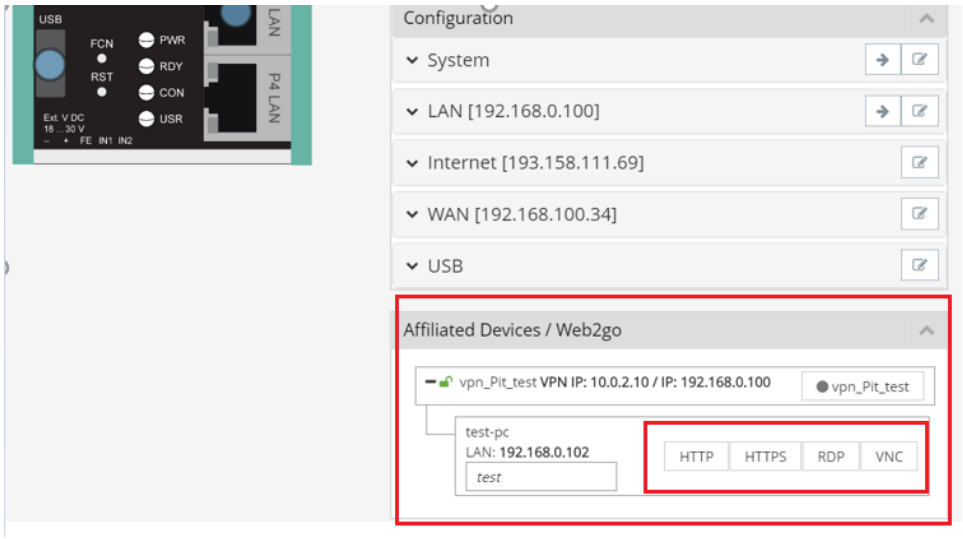
The DRX-IT will set the requested no. for this option:



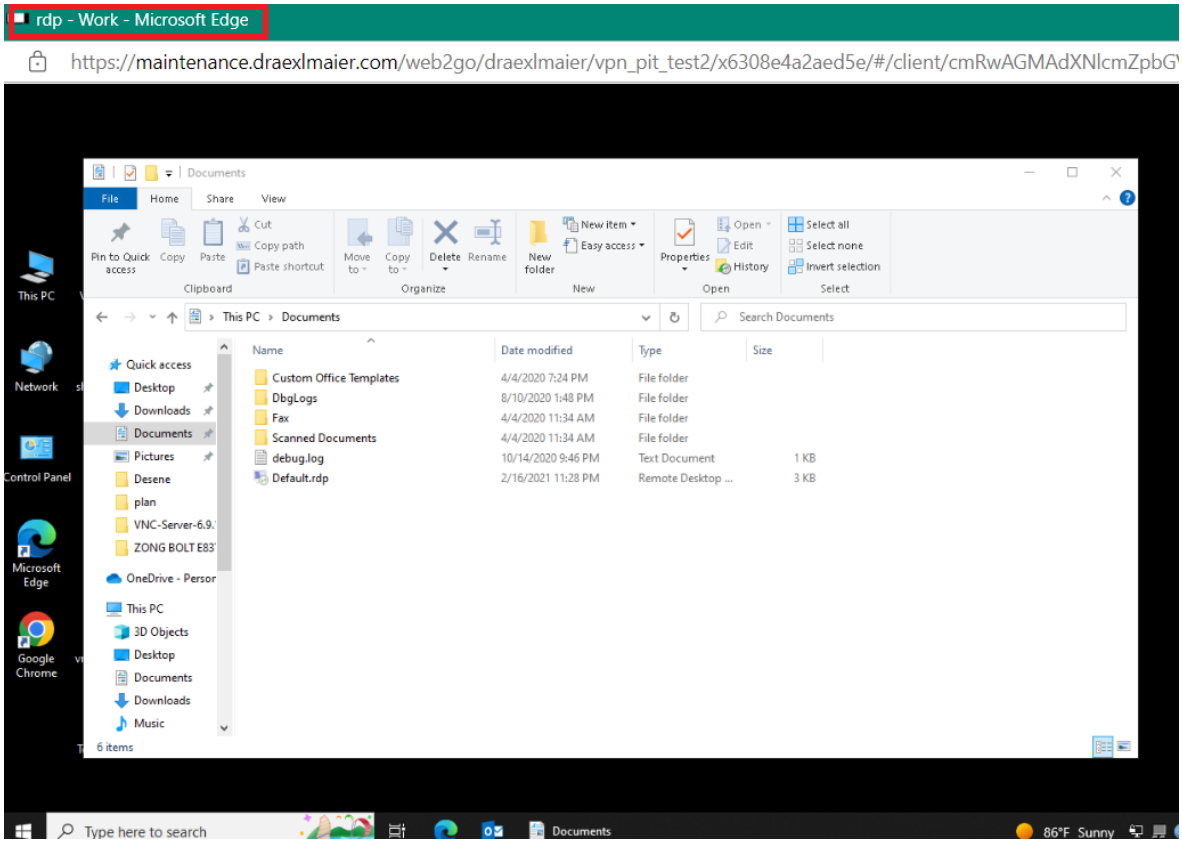
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3.2 Web2go

This is application feature, the user can start a connection Web2go (VNC,RDP,HTTP) without start a vpn tunnel.
Web2go (vnc,rdp, http) should be created by DRX IT if is requested.
Note: Web2go has some limitations. E.g you can't upload data to PLC
Web2go connections have to be requested in doc ***Helmholz_Remote_Support_Template_DRX-accounts_and_devices,*** **only if are needed.**
Open web2go sessions from here:



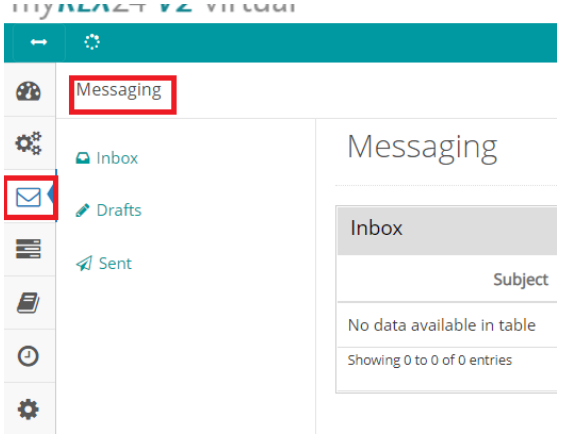
Example Web2go-RDP(web interface):



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3. Messaging

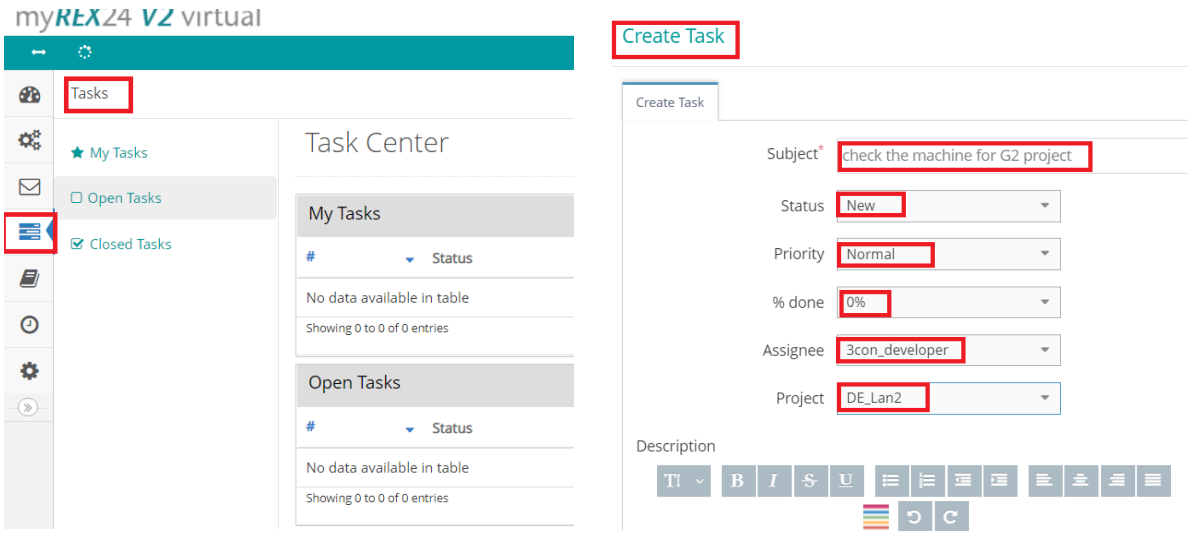
The users can communicate with each other in writing within the portal of virtual machine.



Note:
In addition, under "Notifications" in your user profile, you can specify that you wish to be notified of a new message via e-mail. The address stored in your profile is automatically used as the e-mail address. It is not possible to use an additional or alternative e-mail address.

4. Tasks

You can create tasks and assign them to other users, and to yourself, for processing/completion.



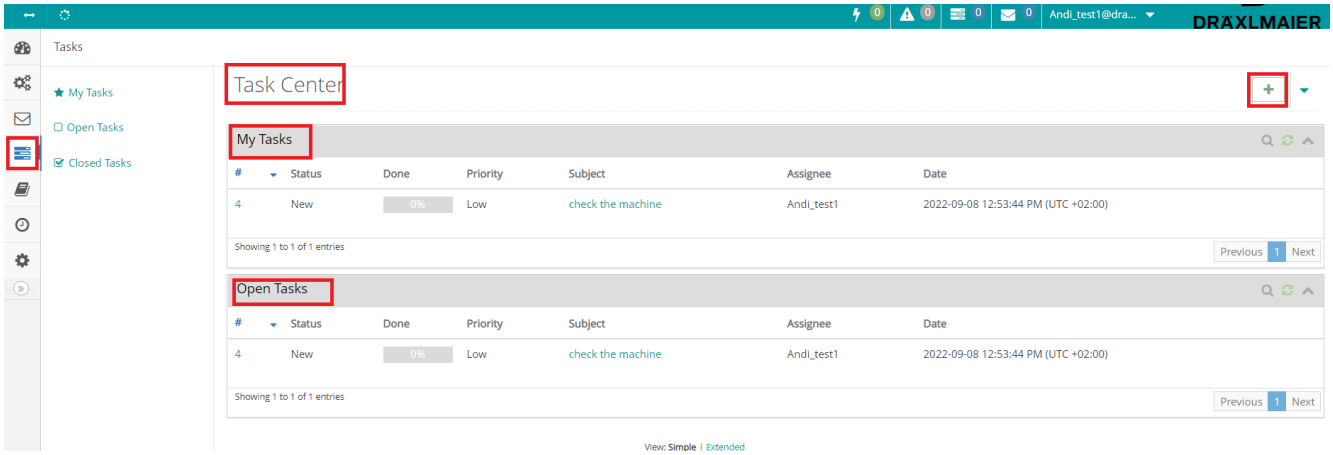
After the creation of the task, you will be forwarded to the "Task Center". Here, you can see the created task in the "Open Tasks" submenu.

At the same time, this is also displayed via the Task icon in the header line  . The number of open tasks

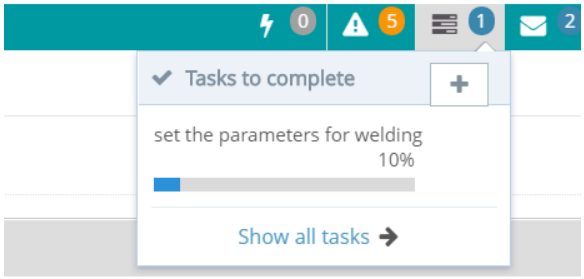
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is output in the LED.
If you have specified it in your user profile, you will receive an e-mail for each task assigned to you.

Task center:

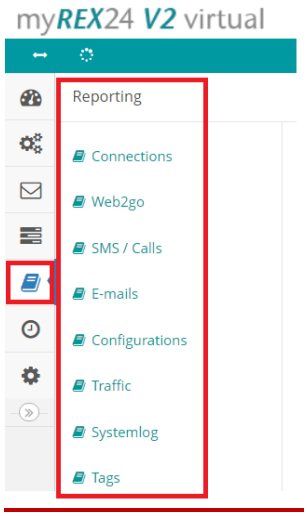


The users tasks:



5. Reporting section

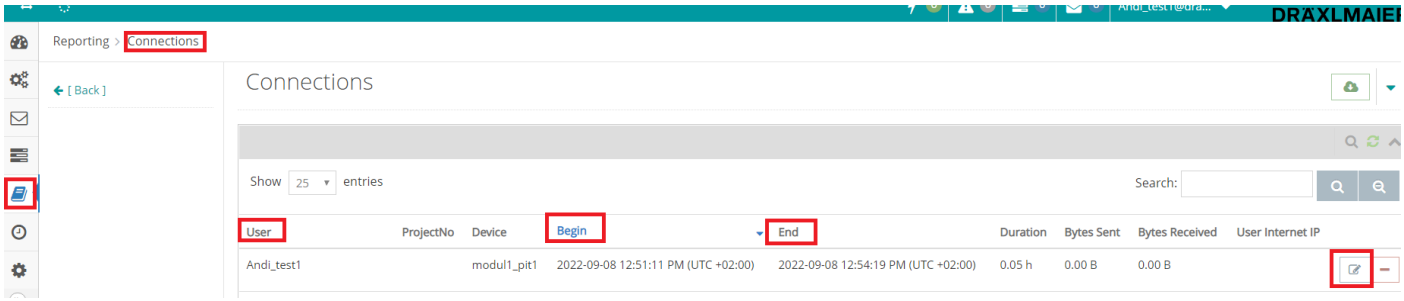
The Remote Service Portal protocols all actions, such as active connections, WEB2go connections, SMS sending, and more. Data-export* can be made in various formats (PDF, HTML, csv) depending on user, device or project-settings.



Connections- List of all active connections.
Web2go- List of all Web2go connections.

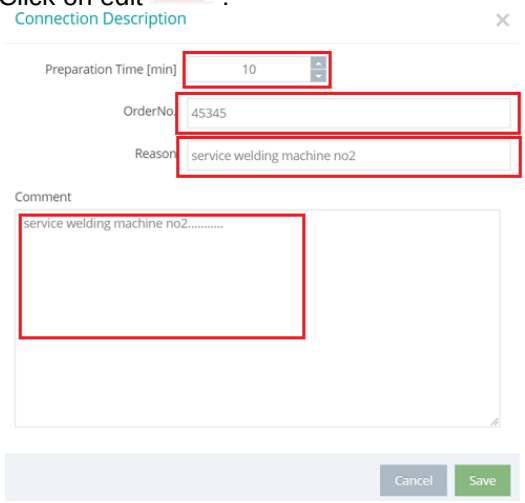
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Note: An active connection only exists when a user has established a connection to a system and activated the VPN tunnel after logging on to the portal. By logging into the portal itself, there is no active connection.



You can add more information about the service connection :

Click on edit  :



6. SEARCHoverIP option – New PLC or device in factory settings

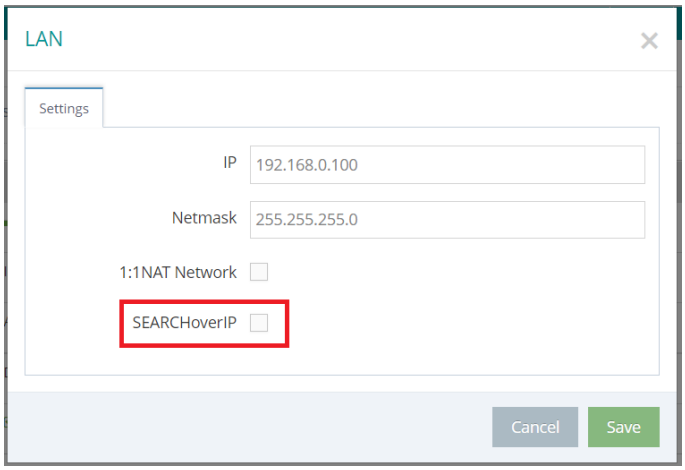
With the **SEARCHoverIP** function, you can also find your PLC via remote maintenance in the network. For example, "virgin" SIEMENS controllers can be found and configured remotely in the network.

Common control systems and their configuration software usually work according to the Broadcast/Multicast/Discover principle (Layer2) during initial commissioning.
The corresponding configuration software searches via network functions the controls in the network. This is usually not a problem in local networks – but in remote maintenance is routed the network (the local network area does not match the remote network area) and therefore only communications over Layer 3 are available. That means that this comfortable way is mostly not available during the remote maintenance and therefore the user has to know the IP - participants and has to specify them manually.

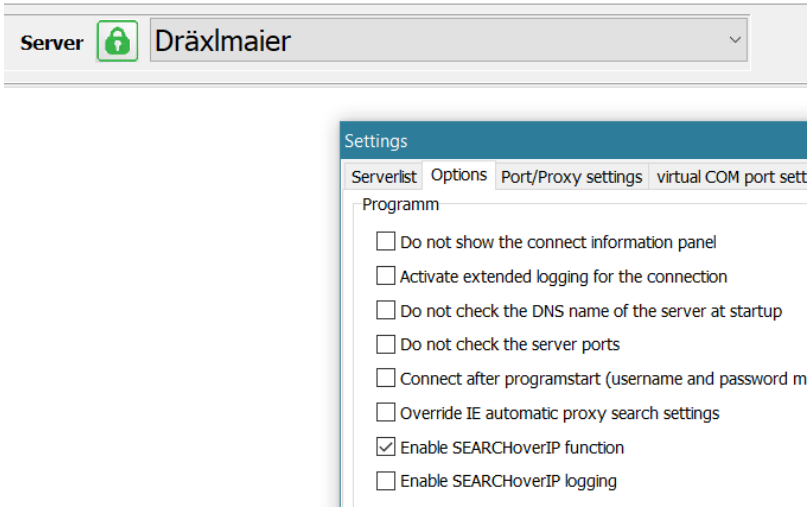
For example: In the Siemens development environment (TIA portal), this search for devices in factory settings is called "Accessible Devices".
For this use case, the "SEARCHoverIP" function should be enabled in the LAN settings of the REX router (for initial commissioning).

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SEARCHoverIP (the option in portal) will be set by Local IT if is requested



In the shDIALUP app this option is activated by default.



SEARCHoverIP option highlighted in portal:



SEARCHoverIP => if the function is activated on the device, SEARCHoverIP appears green highlighted.

SEARCHoverIP => if the function is activated on the device and in the remote client software, the color changes to yellow.
If the function is deactivated in the remote client software, no color changes from green to yellow.
In this case

SEARCHoverIP => Connection setup failed!

- exit your Remote Service Portal session
- restart the remote client software
- log on to the portal again
- connect again to the device (router) in question

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Warning

When the SEARCHoverIP function is activated or executed, the "remote maintenance network" (Remote Client) is connected one to one to the LAN network on the device (router) via Layer2.

- Under certain circumstances, this can lead to IP address conflicts.
- Network disturbances on the remote maintenance side (Remote Client) or router LAN side can occur.

The execution of broadcasts and multicasts on the remote maintenance side (Remote Client) causes an additional increased data traffic through the VPN tunnel.

Notice

Always use the SEARCHoverIP function wisely and as an ultimate ratio.
Activate the function only for the duration of the action to find and configure a controller in order to minimize the above-mentioned risks and to reduce the data traffic in the VPN tunnel.
After completing the work, deactivate the function again in the LAN settings of the router.

A part of devices / programming software that have been tested for use with the SEARCHoverIP function:

Manufacturer	Software	Version	Controller / PLC
Siemens	TIA		S7-1200
Schneider	SoMachine		TM251 MESE
Beckhoff	TwinCat2		Beckhoff TwinCat2
PILZ	PILZ PAS4000		PILZ PAS4000

7. Remote maintenance-TAP Windows Adapter V9-Tia Portal

When connecting additional Ethernet participants (PLC/panel) on the REX 100, it must be ensured that the IP addresses of the devices lie within the address range of the LAN interface of the REX 100.

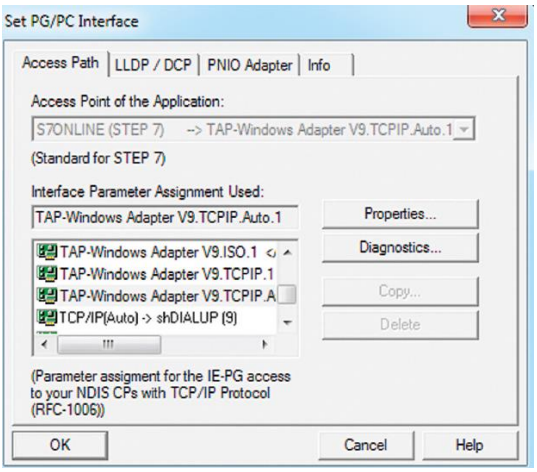
To access a PLC via the VPN tunnel, the OpenVPN adapter must be selected in the corresponding engineering tool (like TIA Portal) as the interface.
In the third-party application (TIA portal for example) you select the “TAP-Windows Adapter V9” network adapter that will be used to get access to the VPN router.

The virtual network connection responsible for this is the **TAP Windows2 Adapter V9:**



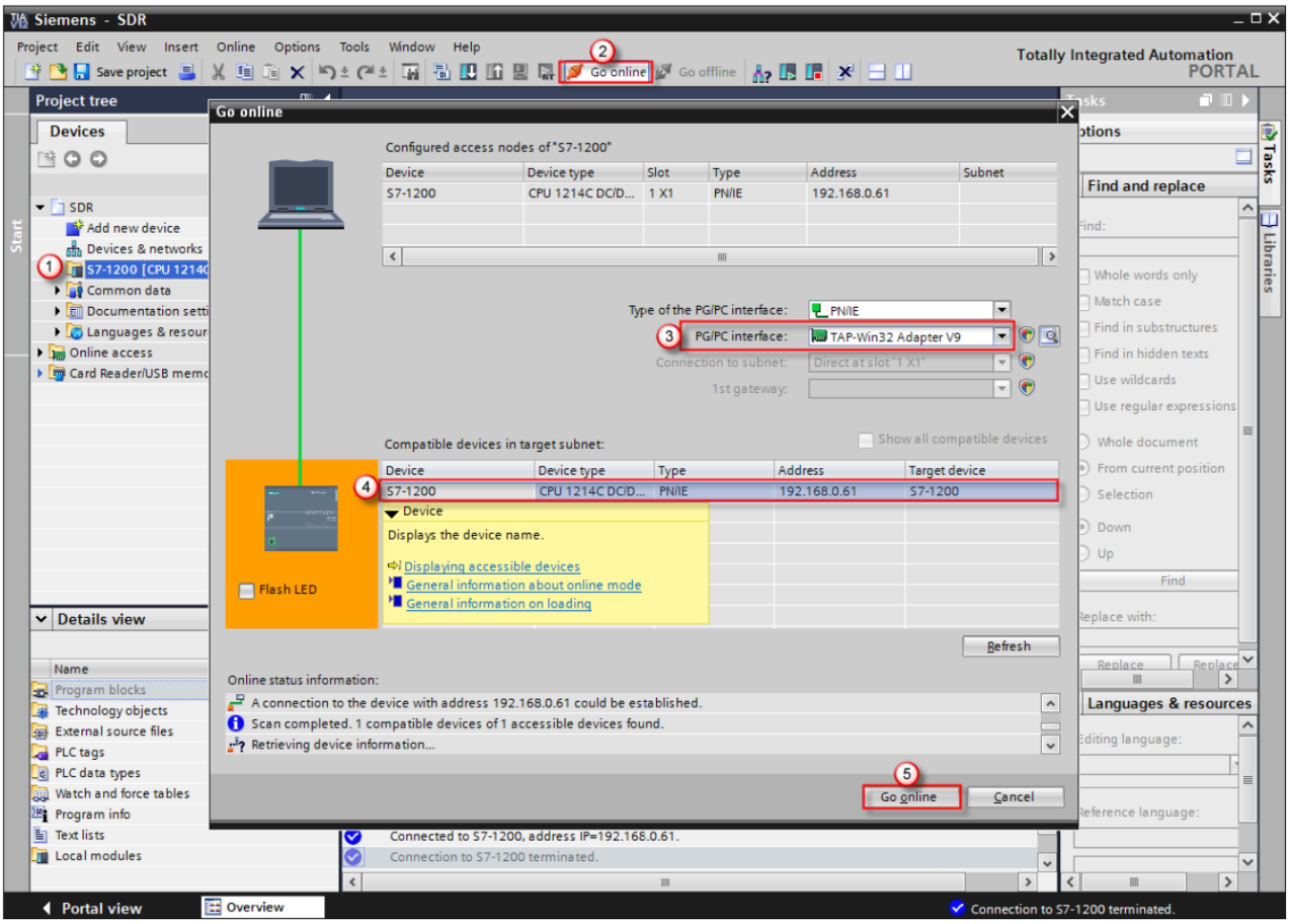
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The setting to be selected in the PG/PC interface module from the SIMATIC STEP 7[1] software



E.g TIA Portal:

The vpn tunnel have to be started.
Open your TIA Portal software.
Open your project and click on Project View at the bottom left of TIA. -1
Select your CPU in the device tree.
Click on the Go Online icon of the toolbar. -2
Select TAP-Win32 Adapter V9 as PG/PC interface. -3
The device is automatically found and appears in the device list . -4
Click the Go Online button to communicate with the detected device. -5
Click Go Offline from the tool bar to disconnect from the device.



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Note: Unfortunately DRÄXLMAIER cannot provide support for 3rd party software. If there are problems with such applications in connection with Helmholz, please use the more direct way and contact the Helmholz support for this:

- Helmholz Service Support: +49 (9135) 7380-0
- Helmholz Ticket System: https://www.helmholz.de/de/support/supportanfrage

Remarks:
"mbNet.mini" devices from MB Connectline are not compatible with the myREX portal. "mbNet" devices (larger variant) from MB Connectline can only be integrated with the "Standard VPN Client" license
The company-wide standard used and also recommended router by the DRÄXLMAIER Group is the following:

• *Helmholz REX100 WAN*

In addition, the following routers are compatible with the RSP (Remote Service Portal):

• *Helmholz REX200 WAN*

• *Helmholz REX250 WAN*

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